Day 3:

Dictionary - { }

Studen = {

"name" : "naveen",

"age : 21,

}

student -> ...

output.

{"name : naveen", "age : 21"}

type(student)

dict

Key call -> Value

name call -> Student

```
Fruit = {
    "apple"=100"
    "Banana"=40
}
print (Fruit ("banana"])
```

output

40

```
Fruit ["banana"] = 28
print ["Fruit"]
```

Output:

["apple' =100, 'banana' =28]

Fruit.pop("apple) -> apple delete
100 no delete

In Function dict:

```
def student_details():
    student ={
      "name" = "Aarthi",
    } "marks" = "100"
    print (student ["name"])
    print (student["marks"])
student_details()
```

Output:

Aarthi
100

Loops:

* A loop is a programming structure that repeats a block of code until a condition become false until All items are processed

why we Use loop:

=> Code repeation ×

=> large Amount of data handle

=> Repeation code Automation.

2 loops:

* For loop (all items to be processed)
* While loop (

For loop:

When we know how many times we want to execute or repeat Something.

Syntax:

For Variable in Sequence → range (or) continuos value

Statement: → what process

ex:

```
For i in range (5)
print (i)
```

range → Build in Function

5 → How many times.

Output:

0
1
2
3
4

For i in range (1, 6) → exclusive
→ Inclusive
print (i)
output:

1
2
3
4
5

→ step size

Range (2, 20, 3)
output:

2
5
8
11
14
17

3 ~~st~~ type Range :
inter...
rep...

1st => How many time run
2nd => 1 to 6
3rd => step size

loop through a list :

Fruits = ["apple", "banana", "orange"]
For Fruit in Fruits :
print (Fruits)

Output:

apple
banana
orange

loop through a string :

name = "python"
For letter in name :
print (letter)

output:

P
Y
T
h
O
N

While loop: Condition Must be.
* Statement
True (or) False
* While loops Runs a longest the Condition is true or the Condition False, Depends upon our Condition.

Syntax:

```
while Condition :
    Statement
```

ex:

```
i=1
while i<=5:
  print(i)
  i+=1        increase by 1
```

Output:

```
1
2
3
4
5
```

ex:

```
while i<=5:
    print(i)
  i=i+1
```

(wrong logic)
(Random Select)

output:
6

Nested loop:
* loop inside loop

ex:

```
For I in range(3):
  For j in range(2):
    print(i, j)
```

Output:

```
0 0
0 1
1 0
1 1
2 0
2 1
```

loop Control:

Break

Continue

Pass ⟶ Build in function

break:

```
For i in range(10):
 if i == 5
  ~~pass~~ break
 print (i)
```

→ break

output:

0
1
2
3
4

Continue:

```
For i in range (5):
 if i == 2
   Continue
 Print (i)
```

→ skip current iteration and balance Continue.

output:

0
1
3
4

Infinite loop:

```
while true:
  print (" hello world ")
```

output

hello world run Many time.

ex:

```
op = [1, 3, 5, 7, 9]
For a in op:
  if a % 5 == 0
  print (F" (a) is divisible by 5")
else
 print (F" (a) is not divisible by 5")
```

Output:

1 is not divisible by 5
3 is not " "
5 is divisible by 5
7 is not divisible by 5
9 is not divisible by 5

```
tup = (1,7,9,6,4,2)
For i in tup:
    if i % 2 == 0:
    print (f"{i} is a even number")
```

output:

6 is a even number
4 " " "
2 " " "

loop through list:

```
Fruits = ["apple", "banana"]
For i in Fruits :
    print ("I like", i)
```

output:

I like apple
I like banana.

Atm pin:

```
attempts = 0
pin_ok = false
while attemps < 3 :
 pin = input (" enter")
 if pin == "1234"
 pin_ok = True
 break
attemps + = 1
else :
    print (" too many attemps")
if pin_ok == True :
 print (" access granted")
```

qn:
* 3 tym wrong password loop exit otherwise login?

Concatenate:

```
str1 = "Hello"
str2 = "world"
result = str1+str2
print(result)
```

(or)

```
str1 = "Hello"
str2 = "world"
result = "".join(str1, str
print (result)
```

```
name = "John"
age = 20
print(F"my name is(name), I am (20)")
```

output:

my name is John, I am 20